5. $f(x) = 2x^2 - 4x + 5 \quad [-1, 3]$

$f(x)$ is continuous and differentiable at $x \in [-1, 3]$

$f(a) = f(-1) = 2\times(-1)^2 - 4\times(-1) + 5 = 2 + 4 + 5 = 11$

$f(b) = f(3) = 2\times 3^2 - 4\times 3 + 5 = 11$

$f(a) = f(b)$, then there is a c in $[-1, 3]$ such that $f'(c) = 0$ $(-1, 3)$

$f'(x) = 4x - 4 = 0$

$x = 1$

At $x = 1$, $f'(1) = 0$

9. $f(x) = 1 - x^{\frac{2}{3}}$

$f'(x) = -\frac{2}{3}x^{-\frac{1}{3}} = -\frac{2}{3\sqrt[3]{x}}$

$f'(x)$ DNE when $x = 0 \Rightarrow f(x)$ is not differentiable at $x = 0$

$\therefore f(x)$ doesn't satisfy the condition of Rolle's rule

25. $f'(x) > 0$ for $x \in [1, 4] \Rightarrow f(x)$ is increasing when $x \in [1, 4]$

$\therefore f(x)$ has a smallest value $f(1) = 0$ when $x \in [1, 4]$

1. $f(x) = x^3 - 5x^2 - 9x + 4$

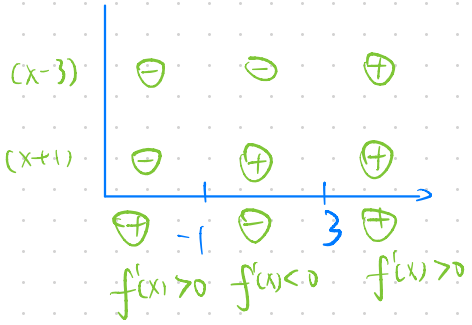
$f'(x) = 3x^2 - 6x - 9$

a) $f'(x) = 3x^2 - 6x - 9 = 0$

$3(x^2 - 2x - 3) = 0$

$3(x-3)(x+1) = 0$

$x = 3$ or $x = -1$



$f'(x) > 0$ when $x < -1$ or $x > 0$

$\Rightarrow f(x)$ is increasing when $x < -1$ or $x > 0$

$f'(x) < 0$ when $-1 < x < 3$

$\Rightarrow f(x)$ is decreasing when $-1 < x < 3$



b) local minimum, $x = 3$

$f(3) = 3^3 - 3 \times 3^2 - 9 \times 3 + 4 = -23$

local maximum, $x = -1$

$f(-1) = (-1)^3 - 3 - 9 \times (-1) + 4 = 9$

c) $f''(x) = 6x - 6 = 0$

$x = 1$

when $x < 1$, $f''(x) < 0$ $\Rightarrow$ concave downward

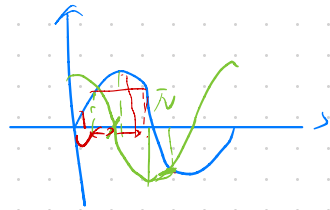
$x > 1$, $f''(x) > 0$ $\Rightarrow$ concave upward.

13. $f(x) = \sin x + \cos x, \quad 0 \le x \le 2\pi$

a) $f'(x) = \cos x - \sin x = 0$

$\cos x = \sin x$

$x = \frac{\pi}{4}$ or $\frac{5\pi}{4}$

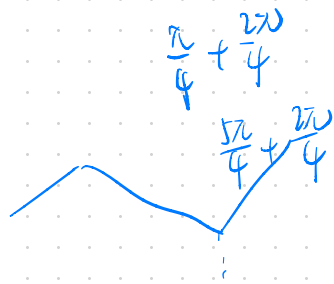


$f'(x) > 0$, when $0 \le x \le \frac{\pi}{4}$ or $\frac{5\pi}{4} \le x \le 2\pi$

$\Rightarrow f(x)$ is increasing ...

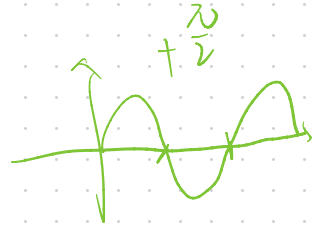
$f'(x) < 0$ when $\frac{\pi}{4} \le x \le \frac{5\pi}{4}$

$\Rightarrow f'(x)$ is decreasing ...



b) At $x = \frac{5\pi}{4}$, local minimum.

$f(\frac{5\pi}{4}) = \sin(\frac{5\pi}{4}) + \cos(\frac{5\pi}{4}) = -1.41$



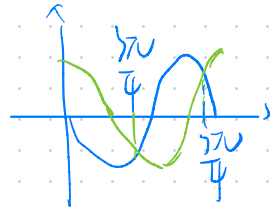
At $x = \frac{\pi}{4}$, local maximum

$f(\frac{\pi}{4}) = \sin(\frac{\pi}{4}) + \cos(\frac{\pi}{4}) = 1.41$

c) $f''(x) = -\sin x - \cos x = 0$

$-\sin x = \cos x$

$x = \frac{3\pi}{4}$ or $x = \frac{7\pi}{4}$



upward

$(0, \frac{3\pi}{4}) \cup (\frac{7\pi}{4}, 2\pi)$

downward

$(\frac{3\pi}{4}, \frac{7\pi}{4})$

$f(x) = \sqrt{x} - \sqrt[4]{x}$ $\quad (x^{\frac{1}{4}})' = \frac{1}{4}x^{-\frac{3}{4}}$

$f'(x) = \frac{1}{2}x^{-\frac{1}{2}} - \frac{1}{4}x^{-\frac{3}{4}} = 0$ $\quad \frac{1}{2\sqrt{x}}$

$\frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{4}x^{-\frac{3}{4}}$

$2x^{-\frac{1}{2}} = x^{-\frac{3}{4}}$, when $x^{-\frac{3}{4}} \neq 0$

$2x^{\frac{1}{4}} = 1$

$x^{\frac{1}{4}} = \frac{1}{2}$ $\quad \frac{1}{\sqrt{x}}$

$x = \frac{1}{16}$ $\quad f(0) = 0$

when $x^{-\frac{3}{4}} = 0 \Rightarrow x = 0$, when $f'(x)$ DNE

So it's singular point

$\therefore x = \frac{1}{16}$ And $x=0$ is endpoint, so $x=0$ can't be local extrema

when $0 < x < \frac{1}{16}$, $f'(x) < 0$, $f(x) \downarrow$

$x > \frac{1}{16}$, $f'(x) > 0$, $f(x) \uparrow$

$\therefore$ local minum $x = \frac{1}{16}$

$f(\frac{1}{16}) = \sqrt{\frac{1}{16}} - \sqrt[4]{\frac{1}{16}} = -0.25$